

PAPER_864: LRC Circuit Pseudo-Monopole Spark-Gap Resonance (1/r Decay)

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****Session:**** 200

****Source:**** advanced_system_analysis_simulator_quantum_calculator.txt (3745 lines)

****Calculator:**** LRCPseudoMonopoleSparkGapResonanceCalc (CP4 #448)

****CVW:**** v2.0.0 compliant

Abstract

An LRC circuit with spark-gap discharge produces a pseudo-monopole magnetic field exhibiting 1/r decay (not the 1/r³ dipole law). Components: L = 75 μH (23 AWG, 10 ft), C = 500 μF (2×1000 μF series), R = 33.3 Ω (3×100 Ω parallel), spark gap 0.5 mm mild steel. The resonance frequency $f_{\text{res}} = 1/(2\pi\sqrt{LC}) = 29.14$ Hz produces $B = 2.53 \times 10^{-8}$ T at 0.61 m with monopole-like 1/r radial falloff, connecting to the UQFF Ug1 Di-Pseudo-Monopole (DPM) field geometry.

1. Core Equations

- $f_{\text{res}} = 1 / (2\pi\sqrt{LC}) = 29.14$ Hz
- $P_{\text{spark}} = E_{\text{spark}} \cdot f_{\text{spark}}$
- $I_{\text{rms}} = \sqrt{2P/R}$
- $B = \mu_0 \cdot I / (2\pi r) = 2.53e-08$ T at 0.61 m
- $B(r) = B_0 \cdot (r_0/r)$ (pseudo-monopole 1/r decay)
- $Q = (1/R) \cdot \sqrt{L/C}$ (quality factor)

2. UQFF Integration

The pseudo-monopole 1/r decay (as opposed to standard dipole 1/r³) is the experimental signature of the Ug1 DPM field geometry at the spark-gap scale. The resonance at 29.14 Hz lies in the ultra-low frequency regime relevant to geomagnetic pulsation coupling. This calculator is differentiated from theoretical DPM classes (PAPER_411 #61, PAPER_855 #439) by its experimental circuit specifications.

3. Source Data

- ****File:**** advanced_system_analysis_simulator_quantum_calculator.txt (3745 lines)
- ****Session:**** 200
- ****Circuit specs:**** L=75μH, C=500μF, R=33.3Ω, spark gap 0.5mm mild steel
- ****VDS/DVP/BH:**** ABSENT

References

1. Murphy, D.T. -- Star Magic UQFF Framework (2024-2026)
2. Biot-Savart law; Maxwell's equations for current loop B-field
3. Dirac, P.A.M. -- Quantised Singularities in the Electromagnetic Field (1931)
4. UQFF Calibration: $\kappa=0.0005/\text{day}$, $[\text{SSq}]=0.57$, $\beta_i \sim 0.603$